

Traffic-Light-Controller-RSLogix -MicroLogix-1100

1. Step-by-Step Explanation (Rung-by-Rung)

Rung 0000 – System Start/Stop

- **I:0/0** (Activation) turns **O:0/0** ON → system active.
- **I:0/1** (Deactivation) turns **O:0/0** OFF → system OFF.
- This establishes a **manual start-stop interlock** for the controller.

Purpose: Enables operator control over when the traffic sequence runs.

Rung 0001 – Default Safety / All Yellow

- Forces all lights to **Yellow** or **No Signal** if system is in a neutral or reset state.
- Output **O:0/9** likely used as “**Yellow for all**” or a warning phase.

Purpose: Ensures fail-safe visual indication (no green conflict).

Rung 0002 – North–South Straight & Right Green

- **Active Timer:** **T4:0** (Preset = 20 s)
- **Condition:** While **T4:0.ACC < 16**, outputs for **NS Green straight & right** are ON.

- When timer reaches 16 s → transitions to next step.
- Outputs: **0:0/1** controls **NS straight/right green**; others are red.

Purpose: Primary North–South flow phase.

Rung 0003 – Yellow Phases

- Uses multiple **GRT (Greater Than)** comparisons:
 - When **T4:0.ACC > 15**, **0:0/2** turns ON → **NS Yellow**.
 - Similar for **T4:3**, **T4:4**, **T4:5** to trigger other direction yellows.
- Outputs:
 - **0:0/6** = NS Left Yellow
 - **0:0/7** = EW Straight Yellow
 - **0:0/8** = EW Left Yellow

Purpose: All yellow transitions between major phases.

Rung 0004 – North–South Left Green

- **Timer:** **T4:3** (Preset = 20 s)
- **Condition:** **T4:3.ACC < 16**
- Output **0:0/3** turns ON → **NS Left Green**, others Red.

Purpose: Left-turn phase for North–South direction.

Rung 0005 – East–West Straight & Right Green

- **Timer:** T4:4 (Preset = 20 s)
- **Condition:** T4:4.ACC between 10–16
- Output 0:0/4 turns ON → **EW Straight & Right Green**, others Red.

Purpose: Main East–West flow.

Rung 0006 – East–West Left Green

- **Timer:** T4:5 (Preset = 20 s)
- **Condition:** T4:5.ACC < 16
- Output 0:0/5 turns ON → **EW Left Green**, others Red.

Purpose: Left-turn phase for East–West direction.

Rung 0007 – END

Marks the end of logic execution.

2. Control Strategy Summary

Control Method

- **Time-based sequential logic**, not state-machine.
- Each major direction and turn uses a dedicated **TON timer** (T4:0–T4:5).
- Transitions controlled by **LES / GRT** comparisons on accumulated timer values.